Annals of Tropical Medicine & Parasitology, Vol. 104, No. 2, 109–121 (2010)

Effects of seasonality and irrigation on malaria transmission in two villages in Côte d'Ivoire

B. G. KOUDOU[*,†,‡], M. DOUMBIA[†], N. JANMOHAMED[§], A. B. TSCHANNEN[†], M. TANNER[¶,§], J. HEMINGWAY[*] and J. UTZINGER[¶,§]

[*]*Vector Group, Liverpool School of Tropical Medicine, Pembroke Place, Liverpool L3 5QA, U.K.*
[†]*Centre Suisse de Recherches Scientifiques, 01 B.P. 1303, Abidjan 01, Côte d'Ivoire*
[‡]*Université d'Abobo-Adjamé, 02 B.P. 801, Abidjan 02, Côte d'Ivoire*
[§]*Centre for Development and Environment, Institute of Geography, University of Bern, CH-3012 Bern, Switzerland*
[¶]*Department of Epidemiology and Public Health, Swiss Tropical and Public Health Institute, P.O. Box, CH-4002 Basel, Switzerland*
[§]*University of Basel, P.O. Box, CH-4003 Basel, Switzerland*

Received 8 January 2010, Revised 21 January 2010, Accepted 25 January 2010

The seasonality of malaria transmission was investigated in two villages in central Côte d'Ivoire: one usually with irrigated rice farming (Zatta) and one without (Tiémélékro). Adult mosquitoes were collected, from February 2002 to August 2005, inside and outside sentinel houses. In Tiémélékro, the biting rate of *Anopheles gambiae* s.s. showed a significant difference between the dry and rainy season only in 2003 ($P<0.001$). The corresponding rates for *An. funestus* s.s. showed significant seasonal differences in both 2002 and 2003 ($P<0.001$ for each year). In Zatta in 2003–2004, when irrigated rice farming was interrupted, there was no significant difference between the *An. gambiae* s.s. biting rates recorded in the dry and rainy seasons. In both 2002 and 2005, however, when irrigated rice farming was practised, the *An. gambiae* s.s. biting rate recorded in Zatta during the rainy season was significantly higher than that seen in the dry season ($P<0.001$ for each year). With just one exception (in Tiémélékro in 2005, the prevalence of *Plasmodium* infection in the *An. funestus* was significantly higher in the rainy season than in the dry season), no significant seasonal differences were seen in the prevalences of *Plasmodium* infection among the *An. gambiae* or *An. funestus*. In conclusion, although malaria transmission is quite stable in central Côte d'Ivoire throughout the year, it can be distinctly modified by irrigated rice farming.

Between 1900 and 2002, social and economic development and control efforts reduced the areas at risk of human malaria by approximately half; from 53% to 27% of the Earth's land surface (Hay *et al.*, 2004). Despite this progress, malaria remains a pressing public-health problem, particularly in sub-Saharan Africa (Greenwood and Mutabingwa, 2002; Hay *et al.*, 2009), where >80% of all malaria-attributable mortality is currently concentrated (Lopez *et al.*, 2006). The transmission of the parasites that cause human malaria (so-called 'malaria transmission') is influenced by myriad environmental factors, including changes in agricultural practices, deforestation, and water-resources development and management (Ijumba and Lindsay, 2001; Ijumba *et al.*, 2002; Keiser *et al.*, 2005; Guerra *et al.*, 2006; Reiter, 2008), climatic factors such as rainfall, humidity and temperature (Reiter, 2008), and various cultural, economic, political and social factors, including health-seeking behaviour, urbanization, armed conflict and war (Essé *et al.*, 2008; Reiter, 2008; Baragatti

Reprint requests to: B. G. Koudou, Vector Group, Liverpool School of Tropical Medicine, Pembroke Place, Liverpool L3 5QA, U.K.
E-mail: g.b.koudou@liv.ac.uk.

© The Liverpool School of Tropical Medicine 2010
DOI: 10.1179/136485910X12607012374154

et al., 2009). Drug resistance in the causative parasites and insecticide resistance in the mosquito vectors are also important factors that now influence malaria transmission (Greenwood *et al.*, 2008; Reiter, 2008). All of these factors may govern the distribution and transmission patterns, including seasonality, of human malaria, depending on the specific features of each endemic setting (Snow, 2000).

Each year, in many parts of Africa, the local populations of anopheline mosquitoes build up rapidly and peak shortly after the onset of the rainy season (Mbogo *et al.*, 1995). In two studies on the relationships between mosquito abundance, malaria transmission and rainfall in West Africa, 70%–90% of the children investigated were found infected with *Plasmodium* spp. after the rainy season (Bonnet *et al.*, 2002; Koudou *et al.*, 2009). It is particularly during and at the end of the rainy season that malaria becomes one of the leading causes of mortality and health-seeking at dispensaries and hospitals in this region (Rey *et al.*, 1987). Not only season but also changing patterns of agriculture, particularly irrigated rice farming, influence malaria transmission in Africa (Ijumba and Lindsay, 2001; Diuk-Wasser *et al.*, 2007; Sogoba *et al.*, 2007). Although the prevalences of human infection with *Plasmodium* and the numbers of clinical cases of malaria are often relatively high in communities living in close proximity to irrigated agriculture, the relationship between malaria transmission and irrigated rice farming is a complex one, since behavioural, cultural and socio–economic factors can all play decisive roles.

In Côte d'Ivoire, human malaria is mostly caused by *Plasmodium falciparum*, with *P. malariae* and *P. ovale* malaria only seen occasionally (Dossou-Yovo *et al.*, 1998; Silué *et al.*, 2008). The main malaria vectors in the country are *Anopheles gambiae* s.s. and members of the *An. funestus* group (Koudou *et al.*, 2005). Recently, malaria transmission, the prevalences of human infection with *Plasmodium*, presumptive malaria and clinically-confirmed malaria, and local knowledge, attitudes, beliefs and practices relating to the prevention and control of 'malaria' have been studied in the villages of Zatta and Tiémélékro, in central Côte d'Ivoire (Koudou *et al.*, 2005, 2007, 2009; Essé *et al.*, 2008). At the time of these studies, Zatta had irrigated rice farming (except in 2003–2004, when a farmers' conflict over land and socio–political issues led to the interruption of such farming in the village) whereas there has never been irrigated rice farming in Tiémélékro. The entomological data from these investigations have now been further analysed, with an emphasis on the effects of seasonality and irrigation on malaria transmission, and form the basis of the present study. The sudden interruption of irrigated rice farming in Zatta, in 2003–2004, offered a unique opportunity to investigate the effect of changing agricultural patterns on malaria transmission.

MATERIALS AND METHODS

Study Area

Details of the study villages [Tiémélékro (6.50°N, 4.17°E) and Zatta (6.86°N, 5.39°E)], the repeated cross-sectional entomological and parasitological surveys that have been conducted, and the investigations of the knowledge, attitudes, beliefs and practices pertaining to 'malaria' have already been published (Girardin *et al.*, 2004; Koudou *et al.*, 2005, 2007, 2009; Essé *et al.*, 2008). In brief, the mean annual temperature is about 28°C and the mean annual precipitation is approximately 1300 mm. There is a long rainy season, from March/April to July, and the 'short rains' occur in October–November. The long dry season runs from December to March and a short dry spell occurs in August and September. Most of the people in both study villages are engaged in subsistence farming.

In Zatta, the irrigated rice farming covers 36 ha (divided into 25 plots), in close

proximity (i.e. within 250 m) to some of the households. In 2002, there were two or even three rice-growing cycles, facilitated by the irrigation system. In 2003 and 2004, irrigation was interrupted because of a farmers' dispute over land property rights and an armed conflict (Koudou *et al.*, 2005, 2007, 2009; Betsi *et al.*, 2006; Fürst *et al.*, 2009) but irrigated rice farming resumed in 2005.

In Tiémélékro, from 2003 to 2005, there was some intensive vegetable (e.g. tomato and okra) farming but, otherwise, subsistence agriculture was the key activity (Girardin *et al.*, 2004).

Throughout the study, in both Tiémélékro and Zatta, preventive measures against malaria, such as sleeping under insecticide-treated nets or indoor residual spraying, were not used or were only used at very low frequency (Essé *et al.*, 2008).

Rainfall

The Ivorian *Société d'Exploitation et de Développement Aéroportuaire, Aéronautique et Météorologique* holds rainfall data for the study area, from 1971 onwards. For the present study, monthly rainfall data for the years 2002, 2003 and 2005 were extracted from the society's records.

Collection of Adult Mosquitoes

Overall, 13 entomological surveys were carried out: seven in the long rainy seasons (in the April and June of 2002, the April, May, June and July of 2003, and the May of 2005), and six in dry seasons (in the February and August of 2002, 2003 and 2005). Adult mosquitoes were collected by means of human-bait night catches. The surveys in 2002 and 2005 were each conducted between 18.00 and 06.00 hours, both inside and outside sentinel houses. Each of the six surveys carried out in 2003, however, covered shorter time periods (from 22:00 to 06:00 hours) and the collectors were only stationed inside the sentinel house because of the unstable socio–political situation at the time. Overall, 96, 48 and 32 night catches were carried out in 2002, 2003 and 2005, respectively. No surveys could be undertaken in 2004, as it was then considered too dangerous to reach the study villages.

Laboratory Procedures

Adult mosquitoes were brought to a laboratory and processed. Firstly, the physiological age of each female *Anopheles* and the corresponding parity 'rate' (i.e. the proportion of female mosquitoes that had laid eggs at least once) were determined by dissection of ovaries and examination of tracheoles (Detinova, 1962). For quality control, a random sample of 10% of the mosquitoes investigated were re-examined by a senior technician. Secondly, each female *Anopheles* was checked for *P. falciparum* infection, in an ELISA for detecting the parasite's circumsporozoite protein (Beier *et al.*, 1988). Thirdly, some of the females belonging to the *An. gambiae* complex were identified to species level, in a PCR-based assay (Scott *et al.*, 1993). Finally, some of the mosquitoes belonging to the *An. funestus* group were further identified using an assay based on a multiplex PCR (Koekemoer *et al.*, 2002; Cohuet *et al.*, 2003).

Ethical Issues

The study protocol was approved by the institutional research commission of the *Centre Suisse de Recherches Scientifiques* (Abidjan, Côte d'Ivoire). Ethical clearance was obtained from the Ivorian Ministry of Public Health and National Malaria Control Programme. The people who acted as bait and collectors in the mosquito collections were all volunteers who signed informed-consent forms. During the study, the collectors were protected against malaria by artesunate–amodiaquine chemoprophylaxis (artesunate–amodiaquine being the recommended, first-line treatment for malaria in Côte d'Ivoire at the time of the present study) and all had been immunized against yellow fever.

Statistical Analysis

The data were double-entered and cross-checked using Excel 2007 (Microsoft) before being analysed using version 9.04 of the STATA software package (Stata Corporation, College Station, TX). Prevalences of *P. falciparum* infection and parity 'rates' in the female *Anopheles* were compared, between the rainy-season and dry-season collections in each collection year, using χ^2 tests. In addition, 95% confidence intervals (CI) were calculated for these prevalences and parity 'rates' assuming binomial distribution. For biting rates, CI were calculated on the basis of a Poisson distribution.

Poisson regression models were fitted to compare the biting rates between the rainy-season and dry-season collections in each collection year, in each of the two study villages. A likelihood-ratio test (LRT) was employed to explore the statistical significance of each of the seasonal differences (rainy season *v.* dry season) seen in the biting rates recorded for *An. gambiae* s.s. and *An. funestus* s.s., in each village in 2002, 2003 and 2005.

In all the data analyses, a *P*-value of <0.05 was considered indicative of a statistically significant difference.

RESULTS

Rainfall and Season

Rainfall patterns in each of the two study villages in 2002, 2003 and 2005 are shown in the Figure. In Zatta, in each study year, the long rainy season lasted from April to July, whilst the short rains ran from mid-September to November. In Tiémélékro, the rainy seasons were somewhat shorter than those seen in Zatta.

Species Composition of *An. gambiae* Complex and *An. funestus* Group

Overall, 110 mosquitoes were identified to species level by PCR: 60 from Tiémélékro and 50 from Zatta. All the 55 tested mosquitoes that were morphologically identified as members of the *An. gambiae* complex were identified as *An. gambiae* s.s. in the PCR-based assay. Similarly, every one of the 55 tested mosquitoes that were considered to be members of the *An. funestus* group was identified as *An. funestus* s.s. using PCR. In further analyses, therefore, all the mosquitoes morphologically identified as members of the *An. gambiae* complex and *An. funestus* group were assumed to be *An. gambiae* s.s. and *An. funestus* s.s., respectively.

Effect of Season on Malaria-vector Behaviour

Table 1 summarizes the number of *An. gambiae* and *An. funestus* caught inside and outside sentinel houses in the study villages, stratified by survey year. Since the methodology of the 2003 mosquito collections (only done inside sentinel houses and for 8 h/collection) differed from that of the other collections, no further details of the 2003 mosquito data are presented here.

In Zatta, 4733 female *An. gambiae* were collected during 96 night catches in 2002, with 60.8% caught during the rainy season, and almost two-thirds collected inside sentinel houses. In 2005, over the 32 night catches, 1137 female *An. gambiae* were collected, with 51.1% caught in the rainy season but only 45.6% caught inside. Although no *An. funestus* were caught in Zatta in 2002, 71 females were collected in 2005, mainly during the dry season and outside sentinel houses.

In Tiémélékro, while more female *An. gambiae* were collected in the dry season of 2002 than in the rainy season (60.9% *v.* 39.1%), the opposite trend was observed in 2005 (34.3% *v.* 65.7%). Moreover, while *An. gambiae* were more frequently collected outside sentinel houses in 2002 than inside, the opposite was observed in 2005. Compared with Zatta, where only a few *An. funestus* were caught (primarily in the dry season of 2005 outside sentinel houses), *An. funestus* were caught in Tiémélékro in

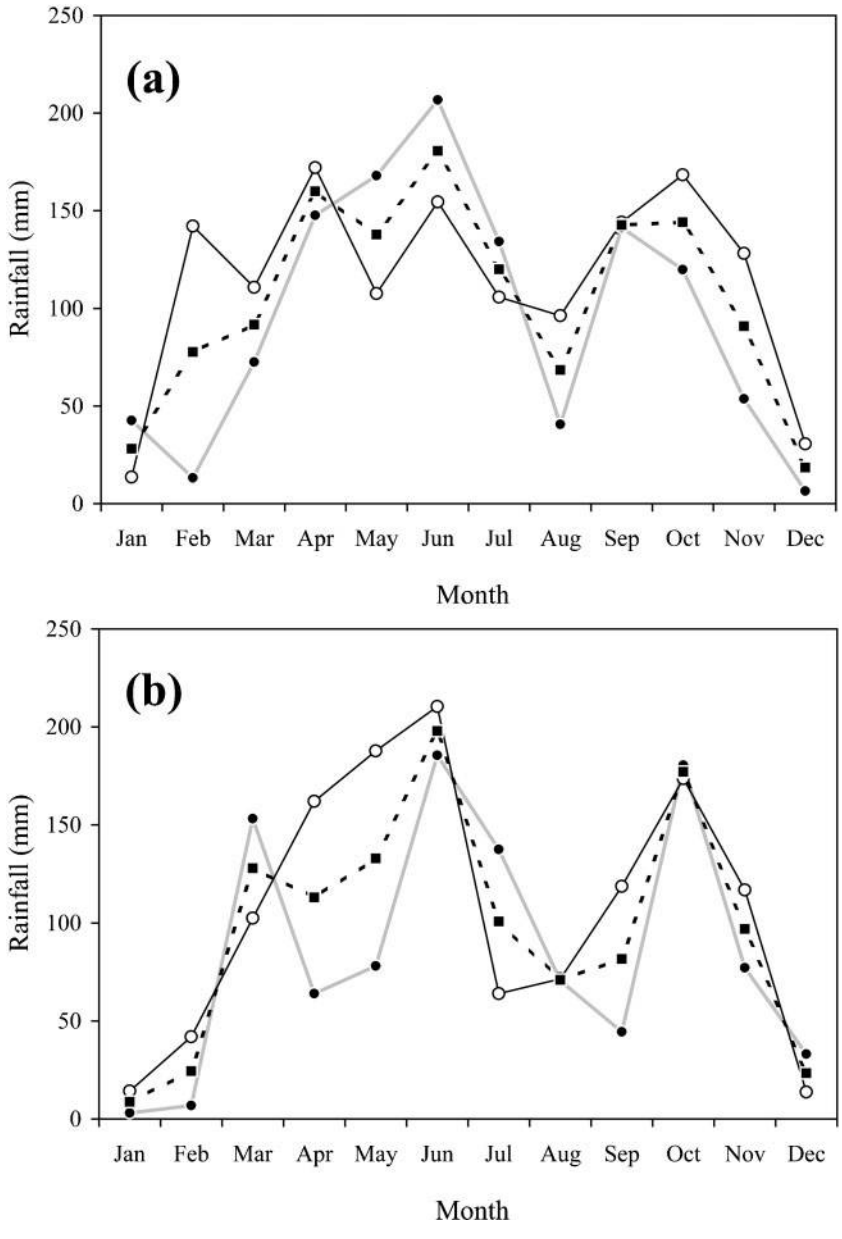


FIG. Monthly rainfall recorded during 2002 (●), 2003 (○) and 2005 (■) in the villages of Zatta (a) and Tiémélékro (b), in central Côte d'Ivoire.

both the rainy and dry seasons and both inside and outside the sentinel houses.

Relationship Between Season and Entomological Parameters

Tables 2 and 3 summarize the mean biting, infection and parity 'rates' and the entomological inoculation rates (EIR) for female *An. gambiae* and *An. funestus* in the two study villages in 2002, 2003 and 2005.

In Zatta, significantly higher *An. gambiae* biting rates were recorded in the dry seasons of 2002 and 2005, when irrigated rice farming was practised, than in the dry

TABLE 1. *The numbers of malaria vectors (i.e. members of the* Anopheles gambiae *complex or* An. funestus *group) caught during the rainy or dry seasons of 2002, 2003 or 2005, in the two study villages in central Côte d'Ivoire*

				No. and (%) of mosquitoes caught:		
Study village	Year	Malaria vector	Season	Inside	Outside	Inside and outside
Zatta	2002	*An. gambiae*	Dry	1139 (24.1)	716 (15.1)	1855 (39.2)
			Rainy	1956 (41.3)	922 (19.5)	2878 (60.8)
			Both	3095 (65.4)	1638 (34.6)	4733 (100.0)
		An. funestus	Dry	0	0	0
			Rainy	0	0	0
			Both	0	0	0
	2003	*An. gambiae*	Dry	114 (30.1)	NA	114 (30.1)
			Rainy	265 (69.9)	NA	265 (69.9)
			Both	379 (100.0)	NA	379 (100.0)
		An. funestus	Dry	22 (37.9)	NA	22 (37.9)
			Rainy	36 (62.1)	NA	36 (62.1)
			Both	58 (100.0)	NA	58 (100.0)
	2005	*An. gambiae*	Dry	246 (21.6)	310 (27.3)	556 (48.9)
			Rainy	273 (24.0)	308 (27.1)	581 (51.1)
			Both	519 (45.6)	618 (54.4)	1137 (100.0)
		An. funestus	Dry	22 (31.0)	43 (60.6)	65 (91.5)
			Rainy	2 (2.8)	4 (5.6)	6 (8.5)
			Both	24 (33.8)	47 (66.2)	71 (100)
Tiémélékro	2002	*An. gambiae*	Dry	441 (28.5)	500 (32.4)	941 (60.9)
			Rainy	277 (17.9)	327 (21.2)	604 (39.1)
			Both	718 (46.5)	827 (53.5)	1545 (100.0)
		An. funestus	Dry	133 (49.3)	103 (38.1)	236 (87.4)
			Rainy	27 (10.0)	7 (2.6)	34 (12.6)
			Both	160 (59.3)	110 (41.1)	270 (100.0)
	2003	*An. gambiae*	Dry	84 (9.6)	NA	84 (9.6)
			Rainy	791 (90.4)	NA	791 (90.4)
			Both	875 (100.0)	NA	875 (100.0)
		An. funestus	Dry	134 (51.3)	NA	134 (51.3)
			Rainy	127 (48.7)	NA	127 (48.7)
			Both	261 (100.0)	NA	261 (100.0)
	2005	*An. gambiae*	Dry	42 (13.7)	63 (20.6)	105 (34.3)
			Rainy	125 (40.8)	76 (24.8)	201 (65.7)
			Both	167 (54.6)	139 (45.4)	306 (100.0)
		An. funestus	Dry	428 (45.2)	507 (53.5)	935 (98.7)
			Rainy	6 (0.6)	6 (0.6)	12 (1.3)
			Both	434 (45.8)	513 (54.2)	947 (100.0)

NA, Not assessed.

season of 2003, when irrigated rice farming was interrupted (with LRT values for the comparison of 2002 with 2003 and the comparison of 2005 with 2003 of 13.79 and 20.50, respectively; $P<0.001$ for each). In 2003, there was no significant seasonal difference in the biting rates of *An. gambiae* (LRT=0.13; $P=0.900$) or *An. funestus* (LRT=0.17; $P=0.879$) in Zatta.

In Tiémélékro, in 2002 (LRT=1.84; $P=0.069$) and 2005 (LRT=0.56; $P=0.455$), there were no significant differences in *An. gambiae* biting rates between the dry and the rainy seasons. In 2003, however, the biting rate was significantly higher in the rainy season than in the dry season (LRT=3.87; $P<0.001$). Regarding *An. funestus* biting rates in Tiémélékro, those recorded in the dry

TABLE 2. *Biting rates, infection 'rates', parity 'rates' and entomological inoculation rates (EIR) recorded among the* Anopheles gambiae *and* An. funestus *during the dry and rainy seasons of 2002, 2003 and 2005, in Zatta, central Côte d'Ivoire*

Malaria vector	Year	Entomological parameter	Mean value, (95% confidence interval) and [no. of mosquitoes investigated] for: Dry season	Rainy season	χ^2 or LRT*	*P*
An. gambiae	2002†	Biting rate (bites/person-season)	38.7 (36.3–41.1)	59.9 (56.9–68.2)	13.79	<0.001
		Infection 'rate' (%)	4.8 (3.6–6.1) [1120]	4.2 (3.0–5.5) [978]	0.33	0.564
		Parity 'rate' (%)	40.4 (36.7–44.1) [673]	31.1 (28.4–33.8) [1157]	16.28	<0.001
		EIR (infective bites/person-season)	338	458		
	2003‡	Biting rate (bites/person-season)	7.1 (5.4–9.3)	8.3 (5.0–11.6)	0.13	0.900
		Infection 'rate' (%)	0.0 (0.0–0.2) [65]	1.7 (0.0–3.6) [176]	1.12	0.290
		Parity 'rate' (%)	61.9 (49.8–74.2) [63]	36.7 (28.6–44.8) [139]	11.16	<0.001
		EIR (infective bites/person-season)	0	26		
	2005†	Biting rate (bites/person-season)	18.3 (11.8–24.7)	58.6 (23.8–93.4)	20.50	<0.001
		Infection 'rate' (%)	2.3 (0.2–5.8) [127]	3.4 (1.7–5.2) [136]	0.37	0.542
		Parity 'rate' (%)	52.9 (45.2–60.6) [194]	46.2 (40.6–52.0) [199]	1.71	0.191
		EIR (infective bites/person-season)	77	365		
An. funestus	2002†	Biting rate (bites/person-season)	0.0 (0.0–0.2)	0.0 (0.0)	0	0
		Infection 'rate' (%)	0.0 (0.0) [0]	0.0 (0.0) [0]	0	0
		Parity 'rate' (%)	0.0 (0.0) [0]	0.0 (0.0) [0]	0	0
		EIR (infective bites/person-season)	0	0		
	2003‡	Biting rate (bites/person-season)	1.4 (0.7–2.4)	1.2 (0.6–2.1)	0.17	0.879
		Infection 'rate' (%)	0.0 (0.0–0.2) [18]	2.3 (0.0–6.8) [44]	0.42	0.519
		Parity 'rate' (%)	0.0 (0.0–0.2) [9]	58.3 (25.6–91.1) [12]	7.88	0.005
		EIR (infective bites/person-season)	0	5		
	2005†	Biting rate (bites/person-season)	2.7 (1.1–4.4)	0.6 (0.0–1.5)	0.61	0.435
		Infection 'rate' (%)	8.3 (0.0–28.4) [8]	0.0 (0.0) [3]	0.87	0.824
		Parity 'rate' (%)	70.9 (24.3–84.3)	60.0 (0.0–100.0)	0.58	0.216
		EIR (infective bites/person-season)	41	0		0

*Showing the results of likelihood-ratio tests (LRT; calculated for biting rates) or χ^2 tests (calculated for infection and parity 'rates') in which the values for the dry and rainy seasons of each study year were compared.

†Irrigated rice farming, performed in a synchronised manner.

‡Irrigated rice cultivation was interrupted in 2003–2004.

TABLE 3. *Biting rates, infection 'rates', parity 'rates' and entomological inoculation rates (EIR) recorded among the* Anopheles gambiae *and* An. funestus *during the dry and rainy seasons of 2002, 2003 and 2005, in Tiémélékro, central Côte d'Ivoire*

Malaria vector	Year	Entomological parameter	Mean value, (95% confidence interval) and [no. of mosquitoes investigated] for: Dry season	Rainy season	χ^2 or LRT*	*P*
An. gambiae	2002†	Biting rate (bites/person-season)	19.6 (18.1–21.1)	12.6 (11.2–14.1)	1.84	0.069
		Infection 'rate' (%)	4.1 (1.7–6.5) [268]	2.6 (1.3–4.0) [531]	1.27	0.260
		Parity 'rate' (%)	72.5 (66.8–78.2) [240]	52.4 (48.4–56.4) [597]	28.34	<0.001
		EIR (infective bites/person-season)	146	60		
	2003‡	Biting rate (bites/person-season)	5.2 (3.8–7.0)	24.7 (21.6–28.3)	3.87	<0.001
		Infection 'rate' (%)	2.6 (0.0–8.7) [35]	7.9 (5.4–10.4) [467]	1.19	0.274
		Parity 'rate' (%)	59.0 (42.8–75.1) [39]	58.6 (54.0–63.1) [449]	0.002	0.961
		EIR (infective bites/person-season)	25	357		
	2005‡	Biting rate (bites/person-season)	3.9 (2.1–5.7)	16.7 (9.1–24.4)	0.56	0.455
		Infection 'rate' (%)	10.9 (3.7–17.3) [52]	6.8 (2.6–10.6) [163]	1.47	0.226
		Parity 'rate' (%)	48.3 (33.3–63.4) [103]	81.1 (75.5–86.7) [201]	30.19	<0.001
		EIR (infective bites/person-season)	77	208		
An. funestus	2002†	Biting rate (bites/person-season)	5.0 (4.1–5.9)	0.7 (0.4–1.1)	6.15	<0.001
		Infection 'rate' (%)	3.1 (0.0–6.6) [97]	7.7 (0.0–18.7) [26]	1.11	0.292
		Parity 'rate' (%)	65.4 (58.5–72.3) [185]	50.0 (27.3–72.7) [22]	2.02	0.155
		EIR (infective bites/person-season)	28	9		
	2003‡	Biting rate (bites/person-season)	8.4 (6.6–10.5)	4.0 (2.9–5.4)	4.50	<0.001
		Infection 'rate' (%)	3.6 (0.0–8.7) [55]	9.1 (1.9–16.2) [66]	1.45	0.229
		Parity 'rate' (%)	75.0 (63.7–86.3) [60]	69.0 (54.4–83.6) [42]	0.44	0.507
		EIR (infective bites/person-season)	55	67		
	2005‡	Biting rate (bites/person-season)	29.9 (20.9–38.9)	1.0 (0.6–1.9)	3.26	0.031
		Infection 'rate' (%)	3.8 (1.2–5.6) [131]	17.6 (0.0–37.8) [11]	4.47	0.035
		Parity 'rate' (%)	65.1 (50.6–73.2) [203]	91.7 (73.3–100.0) [12]	3.61	0.057
		EIR (infective bites/person-season)	207	32		

*Showing the results of likelihood-ratio tests (LRT; calculated for biting rates) or χ^2 tests (calculated for infection and parity 'rates') in which the values for the dry and rainy seasons of each study year were compared.

†Intensive vegetable farming, with two production cycles in the year.

‡Intensive vegetable farming, with just one production cycle in the year.

seasons of 2002 (LRT=6.15) and 2003 (LRT=4.50) were significantly higher than those in the corresponding rainy seasons ($P<0.001$ for each). The difference in the *An. funestus* biting rates between the dry and rainy seasons of 2005 was also statistically significant (LRT=3.26; $P=0.031$).

With regard to infection 'rates', with the exception of the 2005 infection 'rate' for *An. funestus* in Tiémélékro ($\chi^2=4.47$; $P=0.035$), no significant differences were observed between seasons, for *An. gambiae* or *An. funestus*, in either of the study villages (Tables 2 and 3).

In Zatta, although the seasonal variation seen in the parity 'rates' recorded for *An. gambiae* in 2005 was minor ($\chi^2=1.71$; $P=0.191$), the parity 'rates' recorded for this species during the dry seasons in both 2002 ($\chi^2=16.28$; $P<0.001$) and 2003 ($\chi^2=11.16$; $P<0.001$) were significantly higher than those recorded in the corresponding rainy seasons. The parity 'rates' recorded for the *An. funestus* collected in Zatta in 2003 also showed statistically significant differences according to season ($\chi^2=7.88$; $P=0.005$).

In Tiémélékro, no statistically significant, seasonal differences in *An. funestus* parity 'rates' were found in 2002, 2003 or 2005 ($P>0.05$ for each). The parity 'rates' of *An. gambiae* did, however, show such a seasonal difference in both 2002 ($\chi^2=28.34$; $P<0.001$) and 2005 ($\chi^2=30.19$; $P<0.001$).

Malarial Transmission

In both study villages, the EIR for *An. gambiae* were generally higher in the rainy season than in the corresponding dry season. In Zatta, for example, the EIR recorded for *An. gambiae* in the rainy seasons of 2002 and 2005 were 458 and 365 infective bites/person, respectively. In 2003, when irrigated rice farming was interrupted in Zatta, *P. falciparum* transmission by *An. gambiae* and *An. funestus* only probably occurred during the rainy season. In Tiémélékro, in the rainy seasons of 2003 and 2005, the numbers of infective bites/person recorded for *An. gambiae* (357 and 208, respectively) were 14- and three-fold higher than those seen in the corresponding dry seasons (25 and 77, respectively). In 2002, in contrast, the EIR for *An. gambiae* recorded in Tiémélékro in the dry season was 2.5-fold higher than that recorded in the rainy season.

With *An. funestus*, EIR tended to be higher in the dry seasons than in the corresponding rainy seasons, and, in Tiémélékro, this species appeared to be the primary *P. falciparum* transmitter during the dry season of 2005 (when its EIR was 207 infective bites/person).

DISCUSSION

In the present study, the interruption of irrigated rice farming in Zatta — as the result of a farmers' dispute over land property rights and an unstable socio–political situation produced by the 2002–2004 armed conflict in Côte d'Ivoire (Betsi *et al.*, 2006; Fürst *et al.*, 2009) — offered a unique opportunity to study the dynamics of malaria transmission in this village. The present analyses complement those given in previous reports on malaria in the same study villages (Girardin *et al.*, 2004; Koudou *et al.*, 2005, 2007, 2009), now with an explicit focus on the effect of seasonality on malaria transmission under changing agro–ecological conditions. The following points are offered for discussion.

Firstly, many more adult female *An. gambiae* were caught inside sentinel houses in Zatta in years with irrigated rice farming (2002 and 2005) than in 2003 — the study year when no such farming occurred in the village. Since mosquito collection outside the sentinel houses was not feasible in 2003, and there was a somewhat shorter nightly observation period inside sentinel houses in 2003 than in 2002 or 2005, care is needed in the interpretation of these findings. As the peak of *An. gambiae* collection in the study villages occurs around midnight, however

(Koudou *et al.*, 2005), it seems likely that the numbers of *An. gambiae* caught inside the sentinel houses of Zatta in 2003 would not have been much higher if the observation period had been 18.00–06.00 hours (as in 2002 and 2005). Vector abundance in the period of non-irrigated rice farming in Zatta appeared truly lower than when irrigated rice farming was being practised, perhaps because of the reduced numbers of suitable breeding sites, particularly during the dry season. Members of the *An. gambiae* complex are known to explore rice fields when searching for breeding sites (Chandler *et al.*, 1975). Importantly though, even during years of irrigated rice farming, vector abundance showed seasonal variation in Zatta, usually increasing with rainfall, as observed elsewhere (Lindsay *et al.*, 1998; Diuk-Wasser *et al.*, 2005; Bigoga *et al.*, 2007).

Secondly, biting rates of *An. gambiae* in both study villages were usually significantly higher in the rainy season than in the dry season of the same year. When irrigated rice farming was interrupted in Zatta in 2003, much lower biting rates were observed than in the preceding year or in 2005, but there was not a significant seasonal difference in these rates. Hence, the interruption of irrigated rice farming appears to have hidden the 'usual' effect of season on *An. gambiae* biting rates. These findings are in agreement with previous investigations in the humid savannah of Côte d'Ivoire: in an area characterised by intensive agriculture, the biting rate of *An. gambiae* increased significantly a few weeks after the beginning of the rainy season but subsequently decreased and became lowest towards the end of the dry season (Doannio *et al.*, 2006). Moreover, the blunting of seasonal differences in biting rates, as the result of changing patterns in irrigated rice farming, has been documented previously, in the savannahs of both Senegal (Faye *et al.*, 1993) and Mali (Dolo *et al.*, 2004). In the present study, in contrast to the observations on *An. gambiae* (highest biting rates usually observed in the rainy season), the highest biting rates of *An. funestus* were consistently recorded in the dry season, regardless of the prevailing agricultural activity. Moreover, the interruption of the irrigated rice farming in Zatta had no apparent effect on the biting rates of *An. funestus*.

Thirdly, with just one exception, the prevalences of *P. falciparum* (oocyst or sporozoite) infection in both *An. gambiae* and *An. funestus* showed no clear seasonal pattern. In Dielmo, a holo-endemic area in Senegal, however, the infection 'rate' of the vector species of mosquito showed considerable seasonal variation (Fontenille *et al.*, 1997). Similar seasonal variation in mosquito infection has been recorded in the savannah of northern Côte d'Ivoire (Dossou-Yovo *et al.*, 1995), with the prevalence of *P. falciparum* infection in *An. funestus* peaking at the beginning of the dry season in an area with irrigated rice (Dossou-Yovo, 2000). It should be noted, however, that the mean annual infection 'rate' of *An. gambiae* in Zatta was significantly higher when irrigated rice farming was in place (in 2002 and 2005) than in a year (2003) when irrigated rice farming was interrupted (Koudou *et al.*, 2005).

Fourthly, regarding parity 'rates', there was no clear pattern of seasonality for either of the two vector species that were identified. In Tiémélékro, for example, the parity 'rate' for *An. gambiae* was significantly higher in the dry season of 2002 than in that year's rainy season but virtually the same in the dry and rainy seasons of 2003 and significantly lower in the dry season of 2005 than in that year's rainy season.

Fifthly, as discussed elsewhere (Koudou *et al.*, 2005, 2007), the changing patterns of irrigated rice farming in Zatta affected the *An. gambiae*-specific EIR in the village. In brief, the interruption of irrigated rice farming resulted in EIR that were several-fold lower than those seen in 'normal' years. The present results indicate that, despite these changes in the magnitudes of the EIR, the 'usual' seasonal patterns of transmission

remained, with, for example, a considerably higher EIR for *An. gambiae* in Zatta in the rainy season of 2003 than in the dry season of that year (when it dropped to zero). In Tiémélékro, relatively high EIR were recorded throughout the study period but were generally higher in the rainy season than in the corresponding dry season. In this village, *An. funestus* seemed to play an important role in the transmission of *P. falciparum*, particularly in the dry season. The present observations on EIR confirm previous observations made in the northern savannah of Côte d'Ivoire (Dossou-Yovo, 2000) and in southern Cameroon (Bonnet *et al.*, 2002). It is interesting to note that Craig *et al.* (1999), in their use of climatic models, based on monthly rainfall and temperature data, to map areas suitable for malaria transmission in Africa, concluded that a mean of 80 mm of rainfall/month, for at least 3–5 months, was the minimum necessary to enable stable malaria transmission. In southern Cameroon, there is usually a rapid rise in the *An. gambiae* population at the beginning of the short rainy season, followed by an increase in the EIR (Bonnet *et al.*, 2002).

In the present study, the higher *An. funestus* EIR were usually observed during the dry season. Members of the *An. funestus* group, which are often abundant, with a relatively high EIR, during dry seasons (Fontenille *et al.*, 1997; Manga *et al.*, 1997), have been identified as the main malaria vectors in the Guinean climatic region, East Africa and Madagascar (Robert *et al.*, 1985; Severini *et al.*, 1990). As shown in the present study, even in the presence of irrigated rice fields, there is a great variability in the annual EIR, with seasonality apparently playing a key role (Mabaso *et al.*, 2007).

In conclusion, analyses of entomological data revealed that malaria transmission in two different agro–ecological settings of central Côte d'Ivoire is generally very high, albeit with clear seasonal patterns. Whilst the interruption of irrigated rice farming in one of the two study villages resulted in a highly significant reduction in the EIR, the seasonal patterns of transmission remained unchanged. Hence, even in areas of intensive agriculture, the effect of season on malaria transmission must be taken into consideration when designing and monitoring integrated interventions.

ACKNOWLEDGEMENTS. The authors thank the local authorities and villagers of Tiémélékro and Zatta, and the district health officers of Dimbokro and Yamoussoukro, for their commitment in the present study. This investigation received financial support from the *Fonds Ivoiro Suisse de Développement Economique et Sociale* (FISDES), the Swiss Academy of Natural Sciences (SANW) and the Swiss Federal Commission for Fellowships for Foreign Students (CFBEE). While this project was carried out, B.G.K. and M.T. were partially supported by the Swiss National Centre of Competence in Research (NCCR) North–South, via the integrated programme on 'Health and Well-being' (IP4) of the 'Research Partnerships for Mitigating Syndromes of Global Change'. The Swiss National Science Foundation provided financial support to J.U. (via projects PPOOB-102883 and PPOOB-119129).

REFERENCES

Baragatti, M., Fournet, F., Henry, M. C., Assi, S., Ouedraogo, H., Rogier, C. & Salem, G. (2009). Social and environmental malaria risk factors in urban areas of Ouagadougou, Burkina Faso. *Malaria Journal*, **8,** 13.

Beier, M. S., Schwartz, I. K., Beier, J. C., Perkins, P. V., Onyango, F., Koros J. K., Campbell, G. H., Andrysiak, P.M. & Brandling-Bennet, A. D. (1988). Identification of malaria species by ELISA in sporozoite and oocyst infected *Anopheles* from western Kenya. *American Journal of Tropical Medicine and Hygiene*, **39,** 323–327.

Betsi, N. A., Koudou, B. G., Cissé, G., Tschannen, A. B., Pignol, A. M., Ouattara, Y., Madougou, Z., Tanner, M. & Utzinger, J. (2006). Effect of an armed

conflict on human resources and health systems in Côte d'Ivoire: prevention of and care for people with HIV/AIDS. *AIDS Care*, **18,** 356–365.

Bigoga, J. D., Manga, L., Titanji, V. P., Coetzee, M. & Leke, R. G. (2007). Malaria vectors and transmission dynamics in coastal south–western Cameroon. *Malaria Journal*, **6,** 5.

Bonnet, S., Paul, R. E. I., Gouagna, C, Safeukui, I., Meunier, J. Y., Gounoue, R. & Boudin, C. (2002). Level and dynamics of malaria transmission and morbidity in an Equatorial area of south Cameroon. *Tropical Medicine and International Health*, **7,** 249–256.

Chandler, J. A., Highton, R. B. & Hill, M. N. (1975). Mosquitoes of the Kano Plain, Kenya. I. Results of indoor collections in irrigated and nonirrigated areas using human bait and light traps. *Journal of Medical Entomology*, **12,** 504–510.

Cohuet, A., Simard, F., Toto, F., J. C., Kengne, P., Coetzee, M. & Fontenille, D. (2003). Species identification within the *Anopheles funestus* group of malaria vectors in Cameroon and evidence for a new species. *American Journal of Tropical Medicine and Hygiene*, **69,** 200–205.

Craig, M. H., Snow, R. W. & le Sueur, D. (1999). African climatic model of malaria transmission based on monthly rainfall and temperature. *Parasitology Today*, **15,** 105–111.

Detinova, T. S. (1962). *Age-grouping Methods in Diptera of Medical Importance, with Special Reference to Some Vectors of Malaria*. Geneva: World Health Organization.

Diuk-Wasser, M. A., Toure, M. B., Dolo, G., Bagayoko, M., Sogoba, N., Traore, S. F., Manoukis, N. & Taylor, C. E. (2005). Vector abundance and malaria transmission in rice-growing villages in Mali. *American Journal of Tropical Medicine and Hygiene*, **72,** 725–731.

Diuk-Wasser, M. A., Touré, M. B., Dolo, G., Bagayoko, M., Sogoba, N., Sissoko, I., Traoré, S. F. & Taylor, C. E. (2007). Effect of rice cultivation patterns on malaria vector abundance in rice-growing villages in Mali. *American Journal of Tropical Medicine and Hygiene*, **76,** 869–874.

Doannio, J. M. C., Dossou-Yovo, J., Diarrassouba, S., Rakotondraibe, M. E., Chauvancy, G. & Rivière, F. (2006). Comparaison de la composition spécifique et de la dynamique des populations de moustiques dans deux villages du centre de la Côte d'Ivoire, avec et sans périmetre de riziculture irriguée. *Bulletin de la Société de Pathologie Exotique*, **99,** 1–3.

Dolo, G., Briët, O. J. T., Dao, A., Traoré, S. F., Bouaré, M., Sogoba, N., Niaré, O., Bagayogo, M., Sangaré, D., Teuscher, T. & Touré, Y. T. (2004). Malaria transmission in relation to rice cultivation in the irrigated Sahel of Mali. *Acta Tropica*, **89,** 147–159.

Dossou-Yovo, J. (2000). *Etude éthologique des moustiques vecteurs du paludisme en rapport avec les aspects parasitologiques de la transmission du Plasmodium dans la région de Bouaké*. Ph.D. thesis, Université de Cocody, Abidjan, Côte d'Ivoire.

Dossou-Yovo, J., Doannio, J., Rivière, F. & Chauvancy, G. (1995). Malaria in Côte d'Ivoire wet savannah region: the entomological inputs. *Tropical Medicine and Parasitology*, **46,** 263–269.

Dossou-Yovo, J., Ouattara, A., Doannio, J. M. C., Diarrassouba, S. & Chauvancy, G. (1998). Enquêtes paludométriques en zone de savane humide de Côte d'Ivoire. *Médecine Tropicale*, **58,** 51–55.

Essé, C., Utzinger, J., Tschannen, A. B., Raso, G, Pfeiffer, C., Granado, S., Koudou, B. G., N'Goran, E. K., Cissé, G., Girardin, O, Tanner, M. & Obrist, B. (2008). Social and cultural aspects of 'malaria' and its control in central Côte d'Ivoire. *Malaria Journal*, **7,** 224.

Faye, O., Fontenille, D., Hervé, J. P., Diack, P. A., Diallo, S. & Mouchet, J. (1993). Malaria in the sahelian region of Senegal. 1. Entomological transmission findings. *Annales de la Société Belge de Médecine Tropicale*, **73,** 21–30.

Fontenille, D., Lochouarn, L., Diagne, N., Sockhna, C., Lemasson, J. J., Diatta, M., Konaté, L., Faye, F., Rogier, C. & Trape, J. F. (1997). High annual and seasonal variations in malaria transmission by anophelines and vector species composition in Dielmo, a holoendemic area in Senegal. *American Journal of Tropical Medicine and Hygiene*, **56,** 247–253.

Fürst, T., Raso, G., Acka, C. A., Tschannen, A. B., N'Goran, E. K. & Utzinger, J. (2009). Dynamics of socioeconomic risk factors for neglected tropical diseases and malaria in an armed conflict. *PLoS Neglected Tropical Diseases*, **3,** e513.

Girardin, O., Dao, D., Koudou, B. G., Essé, C., Cissé, G., Yao, T., N'Goran, E. K., Tschannen, A. B., Bordmann, G., Lehmann, B., Nsabiman, C., Keiser, J., Killeen, G. F., Singer, B. H., Tanner, M. & Utzinger, J. (2004). Opportunities and limiting factors of intensive vegetable farming in malaria endemic Côte d'Ivoire. *Acta Tropica*, **89,** 109–123.

Greenwood, B. & Mutabingwa, T. (2002). Malaria in 2002. *Nature*, **415,** 670–672.

Greenwood, B. M., Fidock, D. A., Kyle, D. E., Kappe, S. H., Alonso, P. L., Collins, F. H. & Duffy, P. E. (2008). Malaria: progress, perils, and prospects for eradication. *Journal of Clinical Investigation*, **118,** 1266–1276.

Guerra, C. A., Snow, R. W. & Hay, S. I. (2006). A global assessment of closed forests, deforestation and malaria risk. *Annals of Tropical Medicine and Parasitology*, **100,** 189–204.

Hay, S. I., Guerra, C. A., Tatem, A. J., Noor, A. M. & Snow, R. W. (2004). The global distribution and population at risk of malaria: past, present, and future. *Lancet Infectious Diseases*, **4,** 327–336.

Hay, S. I., Guerra, C. A., Gething, P. W., Patil, A. P., Tatem, A. J., Noor, A. M., Kabaria, C. W., Manh, B.

H., Elyazar, I. R., Brooker, S., Smith, D. L., Moyeed, R. A. & Snow, R. W. (2009). A world malaria map: *Plasmodium falciparum* endemicity in 2007. *PLoS Medicine*, **6,** e1000048.

Ijumba, J. N. & Lindsay, S. W. (2001). Impact of irrigation on malaria in Africa: paddies paradox. *Medical and Veterinary Entomology*, **15,** 1–11.

Ijumba, J. N., Mosha, F. W. & Lindsay, S. W. (2002). Malaria transmission risk variations derived from different agricultural practices in an irrigated area of northern Tanzania. *Medical and Veterinary Entomology*, **16,** 28–38.

Keiser, J., Castro, M. C., Maltese, M. F., Bos, R., Tanner, M., Singer, B. H. & Utzinger, J. (2005). Effect of irrigation and large dams on the burden of malaria on a global and regional scale. *American Journal of Tropical Medicine and Hygiene*, **72,** 392–406.

Koekemoer, L. L., Kamau, L, Hunt, R. H. & Coetzee, M. (2002). A cocktail polymerase chain reaction assay to identify members of the *Anopheles funestus* (Diptera: Culicidae) group. *American Journal of Tropical Medicine and Hygiene*, **66,** 804–811.

Koudou, B. G., Tano, Y., Doumbia, M., Nsanzabana, C., Cissé, G., Girardin, O., Dao, D., N'Goran, E. K., Vounatsou, P., Bordmann, G., Keiser, J., Tanner, M. & Utzinger, J. (2005). Malaria transmission dynamics in central Côte d'Ivoire: the influence of changing patterns of irrigated rice agriculture. *Medical and Veterinary Entomology*, **19,** 27–37.

Koudou, B. G., Adja, A. M., Matthys, B., Cissé, G, Koné, M., Tanner, M. & Utzinger, J. (2007). Pratiques agricoles et dynamique de la transmission du paludisme dans deux différentes zones eco–épidemiologiques, au centre de la Côte d'Ivoire. *Bulletin de la Société de Pathologie Exotiques*, **100,** 124–126.

Koudou, B. G., Tano, Y., Keiser, J., Vounatsou, P., Girardin, O., Klero, K., Koné, M., N'Goran, E. K., Cissé, G., Tanner, M. & Utzinger, J. (2009). Effect of agricultural activities on prevalence rates, and clinical and presumptive malaria episodes in central Côte d'Ivoire. *Acta Tropica*, **111,** 268–274.

Lindsay, S. W., Parson, L. & Thomas, C. J. (1998). Mapping the ranges and relative abundance of the two principal African malaria vectors, *Anopheles gambiae sensu stricto* and *An. arabiensis*, using climate data. *Proceedings of the Royal Society of London. Series B, Biological Sciences*, **265,** 847–854.

Lopez, A. D., Mathers, C. D., Ezzati, M., Jamison, D. T. & Murray, C. J. L. (2006). Global and regional burden of disease and risk factors, 2001: systematic analysis of population health data. *Lancet*, **367,** 1747–1757.

Mabaso, M. L., Craig, M., Ross, A., Smith, T. (2007). Environmental predictors of the seasonality of malaria transmission in Africa: the challenge. *American Journal of Tropical Medicine and Hygiene*, **76,** 33–38.

Manga, L., Toto, J. C., Le Goff, G. & Brunhes, J. (1997). The bionomics of *Anopheles funestus* and its role in malaria transmission in a forest area of southern Cameroon. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, **91,** 232–233.

Mbogo, C. N. M., Snow, R. W., Khamala, C. P. M., Kabiru, E. W., Ouma, J. H., Githure, J. L., Marsh, K. & Beier, J. C. (1995). Relationships between *Plasmodium falciparum* transmission by vector populations and the incidence of severe disease at nine sites on the Kenyan coast. *American Journal of Tropical Medicine and Hygiene*, **52,** 201–206.

Reiter, P. (2008). Climate change and mosquito-borne disease: knowing the horse before hitching the cart. *Revue Scientifique et Technique*, **27,** 383–398.

Rey, J. L., Houdier, R., Coulibaly, A. & Soro, B. (1987). Situation du paludisme en Côte d'Ivoire. Résultats préliminaires. *Publications Médicales Africaines*, **78,** 14–18.

Robert, V., Gazin, P., Boudin, C., Molez, J. F., Ouédraogo, V. & Carnevale, P. (1985). La transmission du paludisme en zone de savane arborée et en zone rizicole des environs de Bobo-Dioulasso, Burkina Faso. *Annales de la Société Belge de Médecine Tropicale*, **65** (Suppl. 2), 201–214.

Scott, J. A., Brogdon, W. G. & Collins, F. H. (1993). Identification of single specimens of the *Anopheles gambiae* complex by the polymerase chain reaction. *American Journal of Tropical Medicine and Hygiene*, **49,** 520–529.

Severini, C., Fontenille, D. & Ramiakajato, M. R. (1990). Importance d'*An. funestus* dans la transmission du paludisme au hameau de Mahitsy, à Tananarive, Madagascar. *Bulletin de la Société Pathologie Exotique*, **83,** 114–116.

Silué, K. D., Raso, G., Yapi, A., Vounatsou, P., Tanner, M., N'Goran, E. K. & Utzinger, J. (2008). Spatially-explicit risk profiling of *Plasmodium falciparum* infections at a small scale: a geostatistical modelling approach. *Malaria Journal*, **7,** 111.

Snow, R. W. (2000). The burden of malaria: understanding the balance between immunity, public health and control. *Journal of Medical Microbiology*, **49,** 1053–1055.

Sogoba, N., Doumbia, S., Vounatsou, P., Bagayoko, M. M., Dolo, G., Traoré, S. F., Maïga, H. M., Touré, Y. T. & Smith, T. (2007). Malaria transmission dynamics in Niono, Mali: the effect of the irrigation systems. *Acta Tropica*, **101,** 232–240.